

1. Create a Java application to display student records from database using SELECT query

Code:

```
import java.sql.*;

public class StudentSelect {
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/SIT_CSE";
        String user = "root";
        String password = "root";

        try {
            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish Connection
            Connection con = DriverManager.getConnection(url, user, password);

            System.out.println("Database Connected Successfully!");

            // Create Statement
            Statement st = con.createStatement();

            // SELECT Query
            String query = "SELECT * FROM student";

            // Execute Query
            ResultSet rs = st.executeQuery(query);

            // Display Records
            System.out.println("\n===== STUDENT RECORDS\n=====");

            while (rs.next()) {
                System.out.println("Student ID: " + rs.getInt("Sid"));
                System.out.println("Name: " + rs.getString("Sname"));
                System.out.println("Age: " + rs.getInt("Age"));
                System.out.println("Course: " + rs.getString("Course"));
            }
        }
    }
}
```

```

        System.out.println("-----");
    }

    // Close Connection
    rs.close();
    st.close();
    con.close();

    } catch (Exception e) {
        System.out.println(e);
    }
}

CREATE TABLE student (
    Sid INT PRIMARY KEY,
    Sname VARCHAR(100),
    Age INT,
    Course VARCHAR(50)
);

INSERT INTO student VALUES
(1, 'Akshat', 19, 'CSE'),
(2, 'Rahul', 20, 'CSE'),
(3, 'Aman', 19, 'IT');

```

Output:

```
Database Connected Successfully!

===== STUDENT RECORDS =====
Student ID: 1
Name: Akshat
Age: 19
Course: CSE
-----
Student ID: 2
Name: Rahul
Age: 20
Course: CSE
-----
Student ID: 3
Name: Aman
Age: 19
Course: IT
-----
```

2. Create a Java application that connects to a database and displays product details using a SELECT query. Show product ID, product name, quantity, and price.

Code:

```
import java.sql.*;
```

```
public class ProductSelect {
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/SIT_CSE";
        String user = "root";
        String password = "root";

        try {
            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish Connection
            Connection con = DriverManager.getConnection(url, user, password);

            System.out.println("Database Connected Successfully!");
```

```

// Create Statement
Statement st = con.createStatement();

// SELECT Query
String query = "SELECT * FROM product";

// Execute Query
ResultSet rs = st.executeQuery(query);

// Display Product Details
System.out.println("\n===== PRODUCT DETAILS
=====");

while (rs.next()) {
    System.out.println("Product ID: " + rs.getInt("Pid"));
    System.out.println("Product Name: " + rs.getString("Pname"));
    System.out.println("Quantity: " + rs.getInt("Quantity"));
    System.out.println("Price: " + rs.getDouble("Price"));
    System.out.println("-----");
}

// Close resources
rs.close();
st.close();
con.close();

} catch (Exception e) {
    System.out.println(e);
}
}

CREATE TABLE product (
    Pid INT PRIMARY KEY,
    Pname VARCHAR(100),
    Quantity INT,
    Price DOUBLE
);

INSERT INTO product VALUES

```

```
(101, 'Laptop', 10, 55000),  
(102, 'Mouse', 25, 800),  
(103, 'Keyboard', 15, 1500),  
(104, 'Monitor', 8, 12000);
```

Output:

```
Database Connected Successfully!  
  
===== PRODUCT DETAILS =====  
Product ID: 101  
Product Name: Laptop  
Quantity: 10  
Price: 55000.0  
-----  
Product ID: 102  
Product Name: Mouse  
Quantity: 25  
Price: 800.0  
-----  
Product ID: 103  
Product Name: Keyboard  
Quantity: 15  
Price: 1500.0  
-----  
Product ID: 104  
Product Name: Monitor  
Quantity: 8  
Price: 12000.0  
-----
```